

Deviations of Random Matrices and Geometric Consequences

This chapter is devoted to a remarkably useful uniform deviation inequality for random matrices. Given an $m \times n$ random matrix A , our goal is to show that, with high probability, the approximate equality

$$\|Ax\|_2 \approx \mathbb{E} \|Ax\|_2 \quad (9.1)$$

holds *simultaneously for many vectors* $x \in \mathbb{R}^n$. To quantify how many, we may choose an arbitrary subset $T \subset \mathbb{R}^n$ and ask whether (9.1) holds simultaneously for all $x \in T$. The answer turns out to be remarkably simple: with high probability, we have

$$\|Ax\|_2 = \mathbb{E} \|Ax\|_2 + O(\gamma(T)) \quad \text{for all } x \in T. \quad (9.2)$$

Recall that $\gamma(T)$ is the Gaussian complexity of T , which is a cousin of the Gaussian width we introduced in Section 7.6.2. In Section 9.1 we will deduce the uniform matrix deviation inequality (9.2) from Talagrand's comparison inequality.

The uniform matrix deviation inequality has many consequences. Some are results that we proved earlier by different methods: in Sections 9.2 and 9.3 we will quickly deduce two-sided bounds on random matrices, bounds on random projections of geometric sets, guarantees for covariance estimation for lower-dimensional distributions, and the Johnson–Lindenstrauss lemma and its generalization for infinite sets. New consequences will be proved starting from Section 9.4, where we deduce two classical results about geometric sets in high dimensions: the M^* bound and the escape theorem. Applications to sparse signal recovery will follow in Chapter 10.

9.1 Matrix Deviation Inequality

The following theorem is the main result of this chapter.

Theorem 9.1.1 (Matrix deviation inequality) *Let A be an $m \times n$ matrix whose rows A_i are independent, isotropic, and sub-gaussian random vectors in $\mathbb{R}^n$. Then, for any subset $T \subset \mathbb{R}^n$, we have*

$$\mathbb{E} \sup_{x \in T} \left| \|Ax\|_2 - \sqrt{m} \|x\|_2 \right| \leq CK^2 \gamma(T).$$

Here $\gamma(T)$ is the Gaussian complexity introduced in Section 7.6.2, and $K = \max_i \|A_i\|_{\psi_2}$.

Before we proceed to the proof of this theorem, let us pause to check that $\mathbb{E} \|Ax\|_2 \approx \sqrt{m} \|x\|_2$, so that Theorem 9.1.1 does indeed yield (9.2).

Exercise 9.1.2 (Deviation around expectation) ☕☕ Deduce from Theorem 9.1.1 that

$$\mathbb{E} \sup_{x \in T} \left| \|Ax\|_2 - \mathbb{E} \|Ax\|_2 \right| \leq CK^2 \gamma(T). \quad \blacksquare$$

We will deduce Theorem 9.1.1 from Talagrand's comparison inequality (Corollary 8.6.3). To apply the comparison inequality, all we have to check is that the random process

$$X_x := \|Ax\|_2 - \sqrt{m} \|x\|_2$$

indexed by $x \in \mathbb{R}^n$ has sub-gaussian increments. Let us state this.

Theorem 9.1.3 (Sub-gaussian increments) *Let A be an $m \times n$ matrix whose rows A_i are independent, isotropic, and sub-gaussian random vectors in $\mathbb{R}^n$. Then the random process*

$$X_x := \|Ax\|_2 - \sqrt{m} \|x\|_2$$

has sub-gaussian increments, namely,

$$\|X_x - X_y\|_{\psi_2} \leq CK^2 \|x - y\|_2 \quad \text{for all } x, y \in \mathbb{R}^n. \quad (9.3)$$

Here $K = \max_i \|A_i\|_{\psi_2}$.

Proof of matrix deviation inequality (Theorem 9.1.1) By Theorem 9.1.3 and Talagrand's comparison inequality in the form of Exercise 8.6.4, we get

$$\mathbb{E} \sup_{x \in T} |X_x| \leq CK^2 \gamma(T),$$

as announced. ■

It remains to prove Theorem 9.1.3. Although the proof is a bit longer than most of the arguments in this book, we will make it easier by working out simpler, partial, cases first and gradually moving toward full generality. We develop this argument in the next few subsections.

9.1.1 Theorem 9.1.3 for Unit Vector x and Zero Vector y

Assume that

$$\|x\|_2 = 1 \quad \text{and} \quad y = 0.$$

In this case, the inequality in (9.3) that we want to prove becomes

$$\left\| \|Ax\|_2 - \sqrt{m} \right\|_{\psi_2} \leq CK^2. \quad (9.4)$$

Note that Ax is a random vector in $\mathbb{R}^m$ with independent sub-gaussian coordinates $\langle A_i, x \rangle$, which satisfy $\mathbb{E} \langle A_i, x \rangle^2 = 1$ by isotropy. Then the concentration of the norm theorem 3.1.1 yields (9.4).

9.1.2 Theorem 9.1.3 for Unit Vectors x, y and for a Squared Process

Assume now that

$$\|x\|_2 = \|y\|_2 = 1.$$

In this case, the inequality in (9.3) that we want to prove becomes

$$\left\| \|Ax\|_2 - \|Ay\|_2 \right\|_{\psi_2} \leq CK^2 \|x - y\|_2. \quad (9.5)$$

We first prove a version of this inequality for *squared* Euclidean norms, which are more convenient to handle. Let us guess what form such an inequality should take. We have

$$\begin{aligned} \|Ax\|_2^2 - \|Ay\|_2^2 &= (\|Ax\|_2 + \|Ay\|_2)(\|Ax\|_2 - \|Ay\|_2) \\ &\lesssim \sqrt{m} \|x - y\|_2. \end{aligned} \quad (9.6)$$

The last bound should hold with high probability because the typical magnitude of $\|Ax\|_2$ and $\|Ay\|_2$ is $\sqrt{m}$, by (9.4), and because we expect (9.5) to hold.

Now that we have guessed the inequality (9.6) for the squared process, let us prove it. We are looking to bound the random variable

$$Z := \frac{\|Ax\|_2^2 - \|Ay\|_2^2}{\|x - y\|_2} = \frac{\langle A(x - y), A(x + y) \rangle}{\|x - y\|_2} = \langle Au, Av \rangle, \quad (9.7)$$

where

$$u := \frac{x - y}{\|x - y\|_2} \quad \text{and} \quad v := x + y.$$

The desired bound is

$$|Z| \lesssim \sqrt{m} \quad \text{with high probability.}$$

The coordinates of the vectors Au and Av are $\langle A_i, u \rangle$ and $\langle A_i, v \rangle$. So we can represent Z as a sum of independent random variables:

$$Z = \sum_{i=1}^m \langle A_i, u \rangle \langle A_i, v \rangle.$$

Lemma 9.1.4 *The random variables $\langle A_i, u \rangle \langle A_i, v \rangle$ are independent, mean zero, and sub-exponential; more precisely,*

$$\left\| \langle A_i, u \rangle \langle A_i, v \rangle \right\|_{\psi_1} \leq 2K^2.$$

Proof Independence follows from the construction, but the mean-zero property is less obvious. Although both $\langle A_i, u \rangle$ and $\langle A_i, v \rangle$ do have zero means, these variables are not necessarily independent of each other. Still, we can check that they are uncorrelated. Indeed,

$$\mathbb{E} \langle A_i, x - y \rangle \langle A_i, x + y \rangle = \mathbb{E} (\langle A_i, x \rangle^2 - \langle A_i, y \rangle^2) = 1 - 1 = 0,$$

by isotropy. By the definitions of u and v , this implies that $\mathbb{E} \langle A_i, u \rangle \langle A_i, v \rangle = 0$.

To finish the proof, recall from Lemma 2.7.7 that the product of two sub-gaussian random variables is sub-exponential. So, we get

$$\begin{aligned} \|\langle A_i, u \rangle \langle A_i, v \rangle\|_{\psi_1} &\leq \|\langle A_i, u \rangle\|_{\psi_2} \|\langle A_i, v \rangle\|_{\psi_2} \\ &\leq K\|u\|_2 K\|v\|_2 \quad (\text{by the sub-gaussian assumption}) \\ &\leq 2K^2, \end{aligned}$$

where in the last step we used that $\|u\|_2 = 1$ and $\|v\|_2 \leq \|x\|_2 + \|y\|_2 \leq 2$. $\blacksquare$

To bound Z , we can use Bernstein's inequality (Corollary 2.8.3); recall that it applies for a sum of independent mean-zero sub-exponential random variables.

Exercise 9.1.5 ☕ Apply Bernstein's inequality (Corollary 2.8.3) and simplify the bound. You should get

$$\mathbb{P}\{|Z| \geq s\sqrt{m}\} \leq 2\exp\left(-\frac{cs^2}{K^4}\right)$$

for any $0 \leq s \leq \sqrt{m}$. ☞

Recalling the definition of Z , we can see that we have obtained the desired bound (9.6).

9.1.3 Theorem 9.1.3 for Unit Vectors x, y and for the Original Process

Next, we would like to remove the squares from $\|Ax\|_2^2$ and $\|Ay\|_2^2$ and deduce inequality (9.5) for unit vectors x and y . Let us state this goal again.

Lemma 9.1.6 (Unit y , original process) *Let $x, y \in S^{n-1}$. Then*

$$\left\| \|Ax\|_2 - \|Ay\|_2 \right\|_{\psi_2} \leq CK^2\|x - y\|_2.$$

Proof Fix $s \geq 0$. The conclusion we want to prove is that

$$p(s) := \mathbb{P}\left\{ \frac{|\|Ax\|_2 - \|Ay\|_2|}{\|x - y\|_2} \geq s \right\} \leq 4\exp\left(-\frac{cs^2}{K^4}\right). \quad (9.8)$$

We proceed differently for small and large s .

Case 1: $s \leq 2\sqrt{m}$. In this range we can use our results from the previous subsection. They are stated for the squared process, though. So, to be able to apply those results, we multiply both sides of the inequality defining $p(s)$ by $\|Ax\|_2 + \|Ay\|_2$. With the same Z as we defined in (9.7), this gives

$$p(s) = \mathbb{P}\{|Z| \geq s(\|Ax\|_2 + \|Ay\|_2)\} \leq \mathbb{P}\{|Z| \geq s\|Ax\|_2\}.$$

From (9.4) we know that $\|Ax\|_2 \approx \sqrt{m}$ with high probability. So it makes sense to break the probability that $|Z| \geq s\|Ax\|_2$ into two cases: one where $\|Ax\|_2 \geq \sqrt{m}/2$ and thus $|Z| \geq s\sqrt{m}/2$, and the other where $\|Ax\|_2 < \sqrt{m}/2$ (and then we do not care about Z). This leads to

$$p(s) \leq \mathbb{P} \left\{ |Z| \geq \frac{s\sqrt{m}}{2} \right\} + \mathbb{P} \left\{ \|Ax\|_2 < \frac{\sqrt{m}}{2} \right\} =: p_1(s) + p_2(s).$$

The result of Exercise 9.1.5 gives

$$p_1(s) \leq 2 \exp \left(-\frac{cs^2}{K^4} \right).$$

Further, the bound (9.4) and the triangle inequality give

$$p_2(s) \leq \mathbb{P} \left\{ \left| \|Ax\|_2 - \sqrt{m} \right| > \frac{\sqrt{m}}{2} \right\} \leq 2 \exp \left(-\frac{cs^2}{K^4} \right).$$

Summing the two probabilities, we conclude that the desired bound is

$$p(s) \leq 4 \exp \left(-\frac{cs^2}{K^4} \right).$$

Case 2: $s > 2\sqrt{m}$. Let us look again at the inequality (9.8) that defines $p(s)$ and slightly simplify it. By the triangle inequality, we have

$$\left| \|Ax\|_2 - \|Ay\|_2 \right| \leq \|A(x - y)\|_2.$$

Thus

$$\begin{aligned} p(s) &\leq \mathbb{P} \left\{ \|Au\|_2 \geq s \right\} \quad (\text{where } u := \frac{x - y}{\|x - y\|_2} \text{ as before}) \\ &\leq \mathbb{P} \left\{ \|Au\|_2 - \sqrt{m} \geq s/2 \right\} \quad (\text{since } s > 2\sqrt{m}) \\ &\leq 2 \exp \left(-\frac{cs^2}{K^4} \right) \quad (\text{using (9.4) again}). \end{aligned}$$

Therefore, in both cases we have obtained the desired estimate (9.8). This completes the proof of the lemma. ■

9.1.4 Theorem 9.1.3 in Full Generality

Finally, we are ready to prove (9.3) for arbitrary $x, y \in \mathbb{R}^n$. By scaling, we can assume without loss of generality that

$$\|x\|_2 = 1 \quad \text{and} \quad \|y\|_2 \geq 1. \tag{9.9}$$

(Why?) Consider the contraction of y onto the unit sphere, see Figure 9.1,

$$\bar{y} := \frac{y}{\|y\|_2}. \tag{9.10}$$

Use the triangle inequality to break the increment into two parts:

$$\|X_x - X_y\|_{\psi_2} \leq \|X_x - X_{\bar{y}}\|_{\psi_2} + \|X_{\bar{y}} - X_y\|_{\psi_2}.$$

Since x and $\bar{y}$ are unit vectors, Lemma 9.1.6 may be used to bound the first part. It gives

$$\|X_x - X_{\bar{y}}\|_{\psi_2} \leq CK^2 \|x - \bar{y}\|_2.$$

To bound the second part, note that $\bar{y}$ and y are collinear vectors, so that

$$\|X_{\bar{y}} - X_y\|_{\psi_2} = \|\bar{y} - y\|_2 \|X_{\bar{y}}\|_{\psi_2}.$$

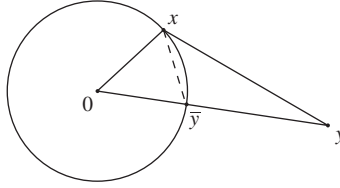


Figure 9.1 Exercise 9.1.7 shows that the triangle inequality can be approximately reversed from these three vectors, and so we have $\|x - \bar{y}\|_2 + \|\bar{y} - y\|_2 \leq \sqrt{2}\|x - y\|_2$.

(Check this identity!) Now, since $\bar{y}$ is a unit vector, (9.4) gives

$$\|X_{\bar{y}}\|_{\psi_2} \leq CK^2.$$

Combining the two parts, we conclude that

$$\|X_x - X_y\|_{\psi_2} \leq CK^2(\|x - \bar{y}\|_2 + \|\bar{y} - y\|_2). \quad (9.11)$$

At this point we might get nervous: we need to bound the right-hand side by $\|x - y\|_2$, but the triangle inequality would give the reverse bound! Nevertheless, looking at Figure 9.1 we may suspect that in our case the triangle inequality can be approximately reversed. The next exercise confirms this rigorously.

Exercise 9.1.7 (Reverse triangle inequality)  Consider vectors $x, y, \bar{y} \in \mathbb{R}^n$ satisfying (9.9) and (9.10). Show that

$$\|x - \bar{y}\|_2 + \|\bar{y} - y\|_2 \leq \sqrt{2}\|x - y\|_2.$$

Using the result of this exercise, deduce from (9.11) the desired bound

$$\|X_x - X_y\|_{\psi_2} \leq CK^2\|x - y\|_2.$$

Theorem 9.1.3 is completely proved. ■

Now that we have proved the matrix deviation inequality (Theorem 9.1.1), we can complement it with a high-probability version.

Exercise 9.1.8 (Matrix deviation inequality: tail bounds)  Show that, under the conditions of Theorem 9.1.1, we have the following. For any $u \geq 0$, the event

$$\sup_{x \in T} \left| \|Ax\|_2 - \sqrt{m}\|x\|_2 \right| \leq CK^2(w(T) + u \operatorname{rad}(T)) \quad (9.12)$$

holds with probability at least $1 - 2 \exp(-u^2)$. Here $\operatorname{rad}(T)$ is the radius of T introduced in (8.47). ■

Exercise 9.1.9  Argue that the right-hand side of (9.12) can be further bounded by $CK^2u\gamma(T)$ for $u \geq 1$. Conclude that the bound in Exercise 9.1.8 implies Theorem 9.1.1.

Exercise 9.1.10 (Deviation of squares)☕☕ Show that, under the conditions of Theorem 9.1.1, we have

$$\mathbb{E} \sup_{x \in T} \left| \|Ax\|_2^2 - m\|x\|_2^2 \right| \leq CK^4 \gamma(T)^2 + CK^2 \sqrt{m} \operatorname{rad}(T) \gamma(T). \quad \blacksquare$$

Exercise 9.1.11 (Deviation of random projections)☕☕☕☕ Prove a version of the matrix deviation inequality (Theorem 9.1.1) for random projections. Let P be an orthogonal projection in $\mathbb{R}^n$ onto an m -dimensional subspace uniformly distributed in the Grassmannian $G_{n,m}$. Show that, for any subset $T \subset \mathbb{R}^n$, we have

$$\mathbb{E} \sup_{x \in T} \left| \|Px\|_2 - \sqrt{\frac{m}{n}} \|x\|_2 \right| \leq \frac{CK^2 \gamma(T)}{\sqrt{n}}.$$

9.2 Random Matrices, Random Projections, and Covariance Estimation

The matrix deviation inequality has a number of important consequences, some of which we present in this and the next section.

9.2.1 Two-Sided Bounds on Random Matrices

First let us apply the matrix deviation inequality to the unit Euclidean sphere $T = S^{n-1}$. In this case, we can recover the bounds on random matrices that we proved in Section 4.6.

Indeed, the radius and Gaussian width of $T = S^{n-1}$ satisfy

$$\operatorname{rad}(T) = 1, \quad w(T) \leq \sqrt{n}.$$

(Recall (7.16).) The matrix deviation inequality in the form of Exercise 9.1.8 together with the triangle inequality imply that the event

$$\sqrt{m} - CK^2(\sqrt{n} + u) \leq \|Ax\|_2 \leq \sqrt{m} + CK^2(\sqrt{n} + u) \quad \forall x \in S^{n-1}$$

holds with probability at least $1 - 2 \exp(-u^2)$.

We can interpret this event as a two-sided bound on the extreme singular values of A (recall (4.5)):

$$\sqrt{m} - CK^2(\sqrt{n} + u) \leq s_n(A) \leq s_1(A) \leq \sqrt{m} + CK^2(\sqrt{n} + u).$$

Thus we recover the result in Theorem 4.6.1.

9.2.2 Sizes of Random Projections of Geometric Sets

Another immediate application of the matrix deviation inequality is the bound on random projections of geometric sets that we gave in Section 7.7. In fact, the matrix deviation inequality yields a sharper bound:

Proposition 9.2.1 (Sizes of random projections of sets) Consider a bounded set $T \subset \mathbb{R}^n$. Let A be an $m \times n$ matrix whose rows A_i are independent, isotropic, and sub-gaussian random vectors in $\mathbb{R}^n$. Then the scaled matrix

$$P := \frac{1}{\sqrt{n}}A$$

(which is a “sub-gaussian projection”) satisfies

$$\mathbb{E} \operatorname{diam}(PT) \leq \sqrt{\frac{m}{n}} \operatorname{diam}(T) + CK^2 w_s(T).$$

Here $w_s(T)$ is the spherical width of T (recall Section 7.5.2) and $K = \max_i \|A_i\|_{\psi_2}$.

Proof Theorem 9.1.1 implies via the triangle inequality that

$$\mathbb{E} \sup_{x \in T} \|Ax\|_2 \leq \sqrt{m} \sup_{x \in T} \|x\|_2 + CK^2 \gamma(T).$$

We can state this inequality in terms of the radii of the sets AT and T as

$$\mathbb{E} \operatorname{rad}(AT) \leq \sqrt{m} \operatorname{rad}(T) + CK^2 \gamma(T).$$

Applying this bound for the difference set $T - T$ instead of T , we can write it as

$$\mathbb{E} \operatorname{diam}(AT) \leq \sqrt{m} \operatorname{diam}(T) + CK^2 w(T).$$

(Here we have used (7.22) to pass from the Gaussian complexity to the Gaussian width.) Dividing both sides by $\sqrt{n}$ completes the proof. $\blacksquare$

Proposition 9.2.1 is more general and sharper than our previous bounds on random projections (Exercise 7.7.3). Indeed, it states that the diameter scales by the exact factor $\sqrt{m/n}$ without an absolute constant in front of it.

Exercise 9.2.2 (Sizes of projections: high-probability bounds)  Use the high-probability version of the matrix deviation inequality (Exercise 9.1.8) to obtain a high-probability version of Proposition 9.2.1. Namely, show that, for $\varepsilon > 0$, the bound

$$\operatorname{diam}(PT) \leq (1 + \varepsilon) \sqrt{\frac{m}{n}} \operatorname{diam}(T) + CK^2 w_s(T)$$

holds with probability at least $1 - \exp(-c\varepsilon^2 m/K^4)$.

Exercise 9.2.3  Deduce a version of Proposition 9.2.1 for the original model of P considered in Section 7.7, i.e., for a random projection P onto a random m -dimensional subspace $E \sim \operatorname{Unif}(G_{n,m})$. 

9.2.3 Covariance Estimation for Lower-Dimensional Distributions

Let us revisit the covariance estimation problem, which we studied in Section 4.7 for sub-gaussian distributions and in Section 5.6 in full generality. We found that the covariance matrix Σ of an n -dimensional distribution can be estimated from $m = O(n)$ sample points for sub-gaussian distributions, and from $m = O(n \log n)$ sample points in full generality.

An even smaller sample can be sufficient for covariance estimation when the distribution is approximately low dimensional, i.e., when $\Sigma^{1/2}$ has low stable rank,¹ which means that

¹ We introduced the notion of stable rank in Section 7.6.1.

the distribution tends to concentrate near a small subspace in $\mathbb{R}^n$. We should expect to do well with $m = O(r)$, where r is the stable rank of $\Sigma^{1/2}$. We noted this only for the general case in Remark 5.6.3, up to a logarithmic oversampling. Now let us address the sub-gaussian case, where no logarithmic oversampling is needed.

The following result extends Theorem 4.7.1 for approximately lower-dimensional distributions.

Theorem 9.2.4 (Covariance estimation for lower-dimensional distributions) *Let X be a sub-gaussian random vector in $\mathbb{R}^n$. More precisely, assume that there exists $K \geq 1$ such that*

$$\|\langle X, x \rangle\|_{\psi_2} \leq K \|\langle X, x \rangle\|_{L^2} \quad \text{for any } x \in \mathbb{R}^n.$$

Then, for every positive integer m , we have

$$\mathbb{E} \|\Sigma_m - \Sigma\| \leq CK^4 \left(\sqrt{\frac{r}{m}} + \frac{r}{m} \right) \|\Sigma\|,$$

where $r = \text{tr } \Sigma / \|\Sigma\|$ is the stable rank of $\Sigma^{1/2}$.

Proof We begin the proof exactly as in Theorem 4.7.1 by bringing the distribution to an isotropic position. Thus,

$$\begin{aligned} \|\Sigma_m - \Sigma\| &= \|\Sigma^{1/2} R_m \Sigma^{1/2}\| \quad \left(\text{where } R_m = \frac{1}{m} \sum_{i=1}^m Z_i Z_i^\top - I_n \right) \\ &= \max_{x \in S^{n-1}} \langle \Sigma^{1/2} R_m \Sigma^{1/2} x, x \rangle \quad (\text{the matrix is positive-semidefinite}) \\ &= \max_{x \in T} \langle R_m x, x \rangle \quad (\text{if we define the ellipsoid } T := \Sigma^{1/2} S^{n-1}) \\ &= \max_{x \in T} \left| \frac{1}{m} \sum_{i=1}^m \langle Z_i, x \rangle^2 - \|x\|_2^2 \right| \quad (\text{by the definition of } R_m) \\ &= \frac{1}{m} \max_{x \in T} \left| \|Ax\|_2^2 - m\|x\|_2^2 \right|, \end{aligned}$$

where in the last step A denotes the $m \times n$ matrix with rows Z_i . As in the proof of Theorem 4.7.1, the Z_i are mean-zero isotropic sub-gaussian random vectors with $\|Z_i\|_{\psi_2} \lesssim 1$. (For simplicity, let us hide the dependence on K within this argument.) This allows us to apply the matrix deviation inequality for A (in the form given in Exercise 9.1.10), which gives

$$\mathbb{E} \|\Sigma_m - \Sigma\| \lesssim \frac{1}{m} (\gamma(T)^2 + \sqrt{m} \text{rad}(T) \gamma(T)).$$

The radius and Gaussian complexity of the ellipsoid $T = \Sigma^{1/2} S^{n-1}$ are easy to compute:

$$\text{rad}(T) = \|\Sigma\|^{1/2} \quad \text{and} \quad \gamma(T) \leq (\text{tr } \Sigma)^{1/2}.$$

(Check!) This gives

$$\mathbb{E} \|\Sigma_m - \Sigma\| \lesssim \frac{1}{m} \left(\text{tr } \Sigma + \sqrt{m} \|\Sigma\| \text{tr } \Sigma \right).$$

Substitute $\text{tr } \Sigma = r \|\Sigma\|$ and simplify the bound to complete the proof. ■

Exercise 9.2.5 (Tail bound) ☕☕☕ Prove a high-probability guarantee for Theorem 9.2.4 that is similar to the results of Exercises 4.7.3 and 5.6.4. Namely, check that, for any $u \geq 0$, we have

$$\|\Sigma_m - \Sigma\| \leq CK^4 \left(\sqrt{\frac{r+u}{m}} + \frac{r+u}{m} \right) \|\Sigma\|$$

with probability at least $1 - 2e^{-u}$.

9.3 The Johnson–Lindenstrauss Lemma for Infinite Sets

Let us now apply the matrix deviation inequality for a finite set T . In this case, we recover the Johnson–Lindenstrauss lemma from Section 5.3, and more.

9.3.1 Recovering the Classical Johnson–Lindenstrauss Lemma

Let us check that the matrix deviation inequality contains the classical Johnson–Lindenstrauss lemma (Theorem 5.3.1). Let $\mathcal{X}$ be a set of N points in $\mathbb{R}^n$ and define T to be the set of normalized differences of $\mathcal{X}$, i.e.,

$$T := \left\{ \frac{x - y}{\|x - y\|_2} : x, y \in \mathcal{X} \text{ are distinct points} \right\}.$$

The Gaussian complexity of T satisfies

$$\gamma(T) \leq C\sqrt{\log N} \quad (9.13)$$

(recall Exercise 7.5.10). The matrix deviation inequality (Theorem 9.1.1) now implies that the bound

$$\sup_{x, y \in \mathcal{X}} \left| \frac{\|Ax - Ay\|_2}{\|x - y\|_2} - \sqrt{m} \right| \lesssim \sqrt{\log N} \quad (9.14)$$

holds with high probability. To keep the calculation simple, we will be satisfied here with a probability 0.99, which can be obtained using Markov’s inequality; Exercise 9.1.8 gives a better probability. Also, for simplicity we have suppressed the dependence on the subgaussian norm K .

Multiply both sides of (9.14) by $(1/\sqrt{m})\|x - y\|_2$ and rearrange the terms. We obtain that, with high probability, the scaled random matrix

$$Q := \frac{1}{\sqrt{m}} A$$

is an approximate isometry on $\mathcal{X}$, i.e.,

$$(1 - \varepsilon)\|x - y\|_2 \leq \|Qx - Qy\|_2 \leq (1 + \varepsilon)\|x - y\|_2 \quad \text{for all } x, y \in \mathcal{X},$$

where

$$\varepsilon \lesssim \sqrt{\frac{\log N}{m}}.$$

Equivalently, if we fix $\varepsilon > 0$ and choose the dimension m such that

$$m \gtrsim \varepsilon^{-2} \log N,$$

then, with high probability, Q is an ε -isometry on $\mathcal{X}$. Thus we recover the classical Johnson–Lindenstrauss lemma (Theorem 5.3.1).

Exercise 9.3.1 🍵 In the argument above, quantify the probability of success and dependence on K . In other words, use the matrix deviation inequality to give an alternative solution of Exercise 5.3.3.

9.3.2 Johnson–Lindenstrauss Lemma for Infinite Sets

The argument above does not really depend on $\mathcal{X}$ being a finite set. We used the fact that $\mathcal{X}$ is finite only to bound the Gaussian complexity in (9.13). This means that we can give a version of the Johnson–Lindenstrauss lemma for general, not necessarily finite, sets. Let us state such a version.

Proposition 9.3.2 (Additive Johnson–Lindenstrauss lemma) *Consider a set $\mathcal{X} \subset \mathbb{R}^n$. Let A be an $m \times n$ matrix whose rows A_i are independent, isotropic, and sub-gaussian random vectors in $\mathbb{R}^n$. Then, with high probability (say, 0.99), the scaled matrix*

$$Q := \frac{1}{\sqrt{m}}A$$

satisfies

$$\|x - y\|_2 - \delta \leq \|Qx - Qy\|_2 \leq \|x - y\|_2 + \delta \quad \text{for all } x, y \in \mathcal{X},$$

where

$$\delta = \frac{CK^2w(\mathcal{X})}{\sqrt{m}}$$

and $K = \max_i \|A_i\|_{\psi_2}$.

Proof Choose T to be the difference set, i.e., $T = \mathcal{X} - \mathcal{X}$, and apply the matrix deviation inequality (Theorem 9.1.1). It follows that, with high probability,

$$\sup_{x, y \in \mathcal{X}} \left| \|Ax - Ay\|_2 - \sqrt{m}\|x - y\|_2 \right| \leq CK^2\gamma(\mathcal{X} - \mathcal{X}) = 2CK^2w(\mathcal{X}).$$

(In the last step we used (7.22).) Dividing both sides by $\sqrt{m}$, we complete the proof. ■

Note that the error δ in Proposition 9.3.2 is additive, while the classical Johnson–Lindenstrauss lemma for finite sets (Theorem 5.3.1) has a multiplicative form of error. This may be a small difference, but in general it is necessary:

Exercise 9.3.3 (Additive error) 🍵 Suppose that a set $\mathcal{X}$ has a non-empty interior. Check that, in order for the conclusion (5.10) of the classical Johnson–Lindenstrauss lemma to hold, one must have $m \geq n$, i.e., no dimension reduction is possible.

Remark 9.3.4 (Stable dimension) The additive version of the Johnson–Lindenstrauss lemma can be stated naturally in terms of the stable dimension of $\mathcal{X}$,

$$d(\mathcal{X}) \sim \frac{w(\mathcal{X})^2}{\text{diam}(\mathcal{X})^2},$$

which we introduced in Section 7.6. To see this, let us fix $\varepsilon > 0$ and choose the dimension m so that it exceeds an appropriate multiple of the stable dimension, namely

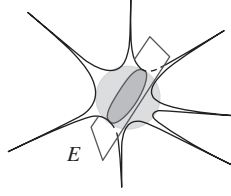


Figure 9.2 Illustration of an M^* bound: the intersection of a set T with a random subspace E .

$$m \geq (CK^4/\varepsilon^2)d(T).$$

Then, in Proposition 9.3.2 we have $\delta \leq \varepsilon \operatorname{diam}(\mathcal{X})$. This means that Q preserves the distances in $\mathcal{X}$ to within a small fraction of the maximal distance, which is the diameter of $\mathcal{X}$.

9.4 Random Sections: M^* Bound and Escape Theorem

Consider a set $T \subset \mathbb{R}^n$ and a random subspace E with given dimension. How large is the typical intersection of T and E ? See Figure 9.2 for an illustration. There are two types of answer to this question. In Section 9.4.1 we give a general bound for the expected diameter of $T \cap E$; it is called the M^* bound. The intersection $T \cap E$ can even be empty; this is the content of the *escape theorem*, which we prove in Section 9.4.2. Both results are consequences of the matrix deviation inequality.

9.4.1 M^* Bound

First, it is convenient to realize the random subspace E as the kernel of a random matrix, i.e., to set

$$E := \ker A,$$

where A is a random $m \times n$ matrix. We always have

$$\dim(E) \geq n - m,$$

and for continuous distributions we have $\dim(E) = n - m$ almost surely.

Example 9.4.1 Suppose that A is a *Gaussian matrix*, i.e., it has independent $N(0, 1)$ entries. Rotation invariance implies that $E = \ker(A)$ is uniformly distributed in the Grassmannian:

$$E \sim \operatorname{Unif}(G_{n, n-m}).$$

Our main result is the following general bound on the diameters of random sections of geometric sets. For historic reasons, this result is called the M^* bound.

Theorem 9.4.2 (M^* bound) *Consider a set $T \subset \mathbb{R}^n$. Let A be an $m \times n$ matrix whose rows A_i are independent, isotropic, and sub-gaussian random vectors in $\mathbb{R}^n$. Then the random subspace $E = \ker A$ satisfies*

$$\mathbb{E} \operatorname{diam}(T \cap E) \leq \frac{CK^2 w(T)}{\sqrt{m}},$$

where $K = \max_i \|A_i\|_{\psi_2}$.

Proof Apply Theorem 9.1.1 for $T - T$ and obtain

$$\mathbb{E} \sup_{x, y \in T} \left| \|Ax - Ay\|_2 - \sqrt{m} \|x - y\|_2 \right| \leq CK^2 \gamma(T - T) = 2CK^2 w(T).$$

If we restrict the supremum to points x, y in the kernel of A then $\|Ax - Ay\|_2$ disappears, since $A(x - y) = 0$, and we have

$$\mathbb{E} \sup_{x, y \in T \cap \ker A} \sqrt{m} \|x - y\|_2 \leq 2CK^2 w(T).$$

Dividing by $\sqrt{m}$ yields

$$\mathbb{E} \operatorname{diam}(T \cap \ker A) \leq \frac{CK^2 w(T)}{\sqrt{m}},$$

which is the bound we claimed. ■

Exercise 9.4.3 (Affine sections) ☕ Check that the M^* bound holds not only for sections through the origin but for all affine sections as well:

$$\mathbb{E} \max_{z \in \mathbb{R}^n} \operatorname{diam}(T \cap E_z) \leq \frac{CK^2 w(T)}{\sqrt{m}},$$

where $E_z = z + \ker A$.

Surprisingly, the random subspace E in the M^* bound is not low dimensional. On the contrary, $\dim(E) \geq n - m$ and we would typically choose $m \ll n$, so E has almost full dimension. This makes the M^* bound a strong and perhaps surprising statement.

Remark 9.4.4 (Stable dimension) It can be enlightening to look at the M^* bound through the lens of the notion of the stable dimension $d(T) \sim w(T)^2 / \operatorname{diam}(T)^2$, which we introduced in Section 7.6. Fix $\varepsilon > 0$. Then the M^* bound can be stated as

$$\mathbb{E} \operatorname{diam}(T \cap E) \leq \varepsilon \operatorname{diam}(T)$$

as long as

$$m \geq C(K^4/\varepsilon^2)d(T). \tag{9.15}$$

In words, *the M^* bound becomes nontrivial – the diameter shrinks – as long as the codimension of E exceeds a multiple of the stable dimension of T .*

Equivalently, the dimension condition states that the sum of the dimension of E and a multiple of the stable dimension of T should be bounded by n . This condition should now make sense from the linear algebraic point of view. For example, if T is a centered Euclidean

ball in some subspace $F \subset \mathbb{R}^n$ then a nontrivial bound $\text{diam}(T \cap E) < \text{diam}(T)$ is possible only if

$$\dim E + \dim F \leq n.$$

(Why?)

Let us look at one remarkable application of the M^* bound.

Example 9.4.5 (The ℓ_1 ball) Let $T = B_1^n$, the unit ball of the ℓ_1 norm in $\mathbb{R}^n$. Since we proved in (7.18) that $w(T) \sim \sqrt{\log n}$, the M^* bound (Theorem 9.4.2) gives

$$\mathbb{E} \text{diam}(T \cap E) \lesssim \sqrt{\frac{\log n}{m}}.$$

For example, if $m = 0.1n$ then

$$\mathbb{E} \text{diam}(T \cap E) \lesssim \sqrt{\frac{\log n}{n}}. \quad (9.16)$$

Comparing this with $\text{diam}(T) = 2$, we see that the diameter shrinks by almost $\sqrt{n}$ as a result of intersecting T with a random subspace E that has almost full dimension (namely, $0.9n$).

For an intuitive explanation of this surprising fact, recall from Section 7.5.4 that the “bulk” of the octahedron $T = B_1^n$ is formed by the inscribed ball $(1/\sqrt{n})B_2^n$. Then it should not be surprising if a random subspace E tends to pass through the bulk and miss the “outliers” that lie closer to the vertices of T . This makes the diameter of $T \cap E$ essentially the same as the size of the bulk, which is $1/\sqrt{n}$.

This example indicates what makes a surprisingly strong and general result such as the M^* bound possible. Intuitively, the random subspace E tends to pass entirely through the bulk of T , which is usually a Euclidean ball with much smaller diameter than T , see Figure 9.2.

Exercise 9.4.6 (M^* bound with high probability) ☕ Use the high-probability version of the matrix deviation inequality (Exercise 9.1.8) to obtain a high-probability version of the M^* bound.

9.4.2 Escape Theorem

In some circumstances, a random subspace E may completely miss a given set T in $\mathbb{R}^n$. This might happen, for example, if T is a subset of the sphere; see Figure 9.3. In this case, the intersection $T \cap E$ is typically empty under essentially the same conditions as in the M^* bound.

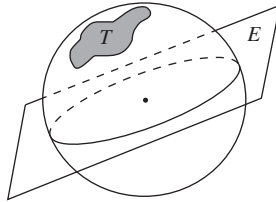


Figure 9.3 Illustration of the escape theorem: the set T has empty intersection with a random subspace E .

Theorem 9.4.7 (Escape theorem) Consider a set $T \subset S^{n-1}$. Let A be an $m \times n$ matrix whose rows A_i are independent, isotropic, and sub-gaussian random vectors in $\mathbb{R}^n$. If

$$m \geq CK^4 w(T)^2 \quad (9.17)$$

then the random subspace $E = \ker A$ satisfies

$$T \cap E = \emptyset$$

with probability at least $1 - 2 \exp(-cm/K^4)$. Here $K = \max_i \|A_i\|_{\psi_2}$.

Proof Let us use the high-probability version of the matrix deviation inequality from Exercise 9.1.8. It states that the bound

$$\sup_{x \in T} \left| \|Ax\|_2 - \sqrt{m} \right| \leq C_1 K^2 (w(T) + u) \quad (9.18)$$

holds with probability at least $1 - 2 \exp(-u^2)$. Suppose that this event indeed holds and $T \cap E \neq \emptyset$. Then for any $x \in T \cap E$ we have $\|Ax\|_2 = 0$, so our bound becomes

$$\sqrt{m} \leq C_1 K^2 (w(T) + u).$$

Choosing $u := \sqrt{m}/(2C_1 K^2)$, we simplify the bound to

$$\sqrt{m} \leq C_1 K^2 w(T) + \frac{\sqrt{m}}{2},$$

which yields

$$\sqrt{m} \leq 2C_1 K^2 w(T).$$

But this contradicts the assumption of the escape theorem, as long as we choose the absolute constant C large enough. This means that the event (9.18) with u chosen as above implies that $T \cap E = \emptyset$. The proof is complete. $\blacksquare$

Exercise 9.4.8 (Sharpness of escape theorem)  Discuss the sharpness of the escape theorem for the example where T is the unit sphere of some subspace of $\mathbb{R}^n$.

Exercise 9.4.9 (Escape from a point set)   Prove the following version of the escape theorem, with a rotation of a point set instead of a random subspace.

Consider a set $T \subset S^{n-1}$ and let $\mathcal{X}$ be a set of N points in $\mathbb{R}^n$. Show that if

$$\sigma_{n-1}(T) < \frac{1}{N}$$

then there exists a rotation $U \in O(n)$ such that

$$T \cap U\mathcal{X} = \emptyset.$$

Here σ_{n-1} denotes the normalized Lebesgue measure (of area) on S^{n-1} . $\blacksquare$

9.5 Notes

The matrix deviation inequality (Theorem 9.1.1) and its proof were borrowed from [128]. Several important related results had been known before. In the partial case where A is Gaussian and T is a subset of the unit sphere, Theorem 9.1.1 can be deduced from Gaussian comparison inequalities. The upper bound on $\|Gx\|_2$ can be derived from the Sudakov–Fernique inequality (Theorem 7.2.11), while the lower bound can be obtained from Gordon’s inequality (Exercise 7.2.14). G. Schechtman [177] proved a version of the matrix deviation inequality in the partial case of Gaussian random matrices A and for general norms (not necessarily Euclidean); we present this result in Section 11.1. For sub-gaussian matrices A , some earlier versions of the matrix deviation inequality can be found in [113, 141, 61]; see [128, Section 3] for a comparison with these results. Finally, a variant of the matrix deviation inequality for sparse matrices A (more precisely, for an A that is a sparse Johnson–Lindenstrauss transform) was obtained in [30].

A version of Proposition 9.2.1 is due to V. Milman [145]; see [11, Proposition 5.7.1]. Theorem 9.2.4 on covariance estimation for lower-dimensional distributions is due to V. Koltchinskii and K. Lounici [115]; they used a different approach that was also based on the majorizing measure theorem. R. van Handel showed in [207] how to derive Theorem 9.2.4 for Gaussian distributions from decoupling, conditioning, and the Slepian lemma. The bound in Theorem 9.2.4 can be reversed [115, 207].

A version of the Johnson–Lindenstrauss lemma for infinite sets that is similar to Proposition 9.3.2 can be found in [128].

The M^* bound, a version of which we proved in Section 9.4.1, is a useful result in geometric functional analysis; see [11, Sections 7.3, 7.4, and 9.3] and [85, 140, 217] for many known variants, proofs, and consequences of M^* bounds. The version we gave here, Theorem 9.4.2, is from [128].

The escape theorem from Section 9.4.2 is also called “escape from the mesh” in the literature. It was originally proved by Y. Gordon [85] for a Gaussian random matrix A and with a sharp constant factor in (9.17). The argument was based on Gordon’s inequality from Exercise 7.2.14. Matching lower bounds for this sharp theorem are known for spherically convex sets [184, 9]. In fact, for a spherically convex set, the exact value of the hitting probability can be obtained by the methods of integral geometry [9]. Oymak and Tropp [159] proved that this sharp result is universal, i.e., it can be extended to non-Gaussian matrices. Our version of the escape theorem (Theorem 9.4.7), which is valid for even more general classes of random matrices but does not feature sharp absolute constants, is borrowed from [128]. As we will see in Section 10.5.1, the escape theorem is an important tool for signal recovery problems.